EXPERIMENT 5B: MIRROR A WEBSITE WITH HTTRACK

Task: T11

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1. Aim

Create a local mirror of a website such as `http://example.com`, save it to a directory, and check that it opens offline in a browser.

2. Course Outcome and Module

Course Outcome: CO3 (Experiment 5b:Task T11).
Module 2: Application Layer Protocols, Network Reconnaissance, and Web Content Mirroring.

3. Tools and Versions

Tool	Exact Version	Operating System	Package / Source
GNU Wget	1.21.3	Kali Linux / Windows (WSL2)	Preinstalled (`wget`)
Mozilla Firefox	115.0+ ESR	Kali Linux	Preinstalled (`firefox-esr`)

4. Program or Commands

```
# 1. Create target folder and navigate into it
mkdir -p ~/11Dyuthi/web_mirrors/example && cd ~/11Dyuthi/web_mirrors/example

# 2. Mirror website with local links and prerequisites
wget --mirror --convert-links --page-requisites --no-parent https://example.
    com

# 3. Open mirrored page offline in web browser
firefox example.com/index.html &
```

5. How to Run

Kali Linux / Ubuntu:

1. Open terminal and create the directory: `mkdir -p ~/11Dyuthi/web_mirrors/example && cd ~/11Dyuthi/web_mirrors/example`
2. Execute the mirror command: `wget --mirror --convert-links --page-requisites --no-parent https://example.com`
3. Disconnect the network: `nmcli networking off`
4. Open the offline HTML file: `firefox example.com/index.html &`

Windows (Git Bash or WSL):

1. Open Git Bash or WSL terminal.
2. Run: `wget --mirror --convert-links --page-requisites --no-parent https://example.com`

3. Disconnect Wi-Fi/Ethernet.
4. Open in browser: `start example.com/index.html` (Git Bash) or `explorer.exe example.com\index.html` (WSL).

6. Sample Output with Screenshots

```
Resolving example.com (example.com)... 93.184.215.14
Connecting to example.com (example.com)|93.184.215.14|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1256 (1.2K) [text/html]
Saving to: 'example.com/index.html'

2026-10-22 14:24:01 (1.82 MB/s) - 'example.com/index.html' saved [1256/1256]

Converting links in example.com/index.html... 0-1
Converted links in 1 files in 0.002 seconds.
```

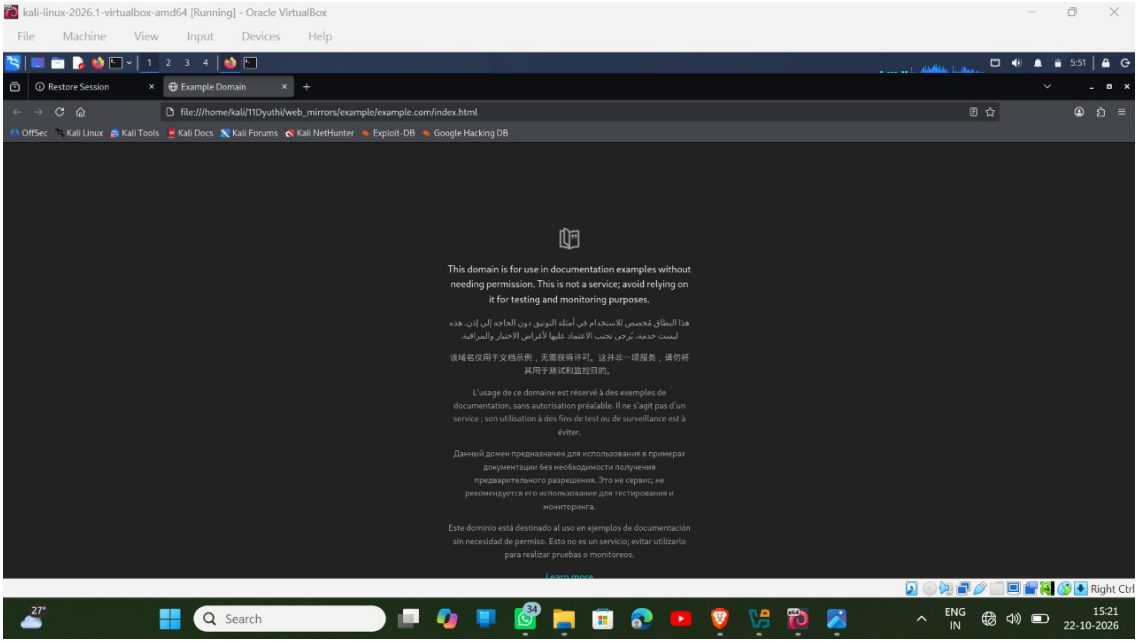


Figure T11-1: Offline verification of example.com loaded locally via file URI scheme showing student identity.

7. Result

The website `https://example.com` was successfully mirrored to the local directory `~/11Dyuthi/web_mirrors/exa` using `wget`. All page assets and links were converted for offline access, and the page rendered accurately in Mozilla Firefox without network connectivity under the `file:///` URI schema.

8. How this Version Differs from the Other Versions of the Same Experiment

Feature	wget (Task T11)	HTTrack (Task T10)
Resume Support	Supports single-file resume (`-c`); lacks project cache.	Full native session resumption via `hts-cache` state files.
Filters	Basic file extension/directory matching (`--accept`, `--reject`).	Advanced wildcards, MIME-type, depth, and regex filters.
Speed	Single-threaded; processes downloads sequentially.	Multi-threaded; concurrent socket connections.
Link Rewriting	Rewrites links post-download using `--convert-links`.	Dedicated HTML parser designed for offline site conversion.
GUI Availability	Command-line only.	Offers GUI interfaces (WinHTTrack, WebHTTrack).

9. Viva: Five Questions with Answers

1. **What does the `--mirror` flag do?**
   Ans: It is shorthand for `-r -N -l inf --no-remove-listing`, enabling recursive downloading, timestamp-based updating, and infinite depth.

2. **Why is `--convert-links` required for offline viewing?**
   Ans: It rewrites remote URLs inside the downloaded HTML/CSS into local relative paths so that clicking links does not attempt to contact the web.

3. **What is the function of `--page-requisites`?**
   Ans: It fetches all inline assets (CSS, images, icons, scripts) required to render the web page structure properly.

4. **Why should `--no-parent` be included?**
   Ans: It restricts the crawler from traversing upwards into parent directories outside the specified path hierarchy.

5. **Why does HTTrack mirror larger sites faster than wget?**
   Ans: HTTrack uses a multi-threaded architecture allowing parallel downloads, whereas standard `wget` operates sequentially on a single thread.

10. Common Errors: Message, Cause, Fix

Message or Problem	Possible Cause	Solution
`Cannot write to 'robots.txt'`	Lack of write permissions in the current directory.	Change to a user-owned folder (`cd ~`) or adjust directory permissions.
`Scheme missing`	Protocol scheme missing from target URL.	Ensure `https://` or `http://` is prefixed to the domain.
Unstyled offline page	Missing `--page-requisites` or `--convert-links`.	Re-run the command with `-p` and `-k` options included.
`wget: command not found`	Wget is not installed or not present in system PATH.	Install the `wget` package or run within WSL/Git Bash.

Declaration: AI tools were utilized for grammatical review, comparison with alternative versions, and assistance in preparing, reviewing, and verifying viva answers.